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Fair face forgery detection via cross-domain decoupling and entropy-adaptive enhancement

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ABSTRACT

Recent advances in generative face forgery detection have improved detection performance, yet growing evidence shows these models often exhibit unfair disparities across demographic groups, such as race and gender. Such disparities may lead to certain groups being disproportionately misclassified or excluded from protection, ultimately undermining trust and adoption of detection technologies. This reveals an urgent need to develop methods that jointly ensure both fairness and generalization in face forgery detection. Current approaches based on spatial-domain decoupling promote fairness but often lack generalization, whereas frequency-domain decoupling improves generalization but compromises fairness. To address this challenge, we propose a novel fair generalization learning framework. Our approach disentangles domain-invariant and high-frequency forgery features through cross-domain decoupling, and performs fairness-aware generalization via a carefully designed cross-domain attention fusion mechanism. We introduce two frequency-domain auxiliary tasks — a feature perception task and a feature contrastive learning task — to guide the learning of discriminative frequency-domain representations. Furthermore, we incorporate an entropy-weighted adaptive enhancement strategy to dynamically improve both fairness and generalization. Experiments on multiple benchmark deepfake datasets demonstrate that our method consistently outperforms state-of-the-art baselines in balancing fairness and generalization.

1. Introduction

The rapid advancement of deep generative models, such as generative adversarial networks (GANs) [1] and diffusion models [2], has profoundly transformed facial media synthesis. These powerful techniques now enable photorealistic face manipulation with remarkable fidelity, generating synthetic content that is often indistinguishable from real human appearances even to experts [3]. Such capabilities have raised significant concerns regarding societal security, including privacy breaches [4], financial fraud [5], and other malicious applications. These emerging threats have highlighted the urgent need for robust research on face forgery detection. Mainstream face forgery detection methods leverage various forgery cues — including biological signals [6], blending artifacts [7], and frequency-domain signals [8] — for detection tasks. Although these methods have achieved strong generalization performance, recent studies and reports [9] have uncovered fairness issues in current forgery detection models. For example, some state-of-the-art detectors exhibit higher accuracy when evaluating deepfakes of individuals with lighter skin tones than those with darker skin tones [9], and they also achieve higher accuracy on male samples than on female ones [10]. In cross-dataset scenarios, some fairness optimization methods suffer from overall performance degradation on unseen domains, accompanied by an increased disparity in misclassification rates across different demographic groups. This critical flaw allows attackers to generate malicious deepfakes targeting specific populations to evade detection. Therefore, reconciling the trade-off between fairness and generalization remains a fundamental challenge. Existing methods still cannot effectively address this challenge.

In face forgery detection, existing studies focus on either fairness or generalization and fall into two main categories, both of which

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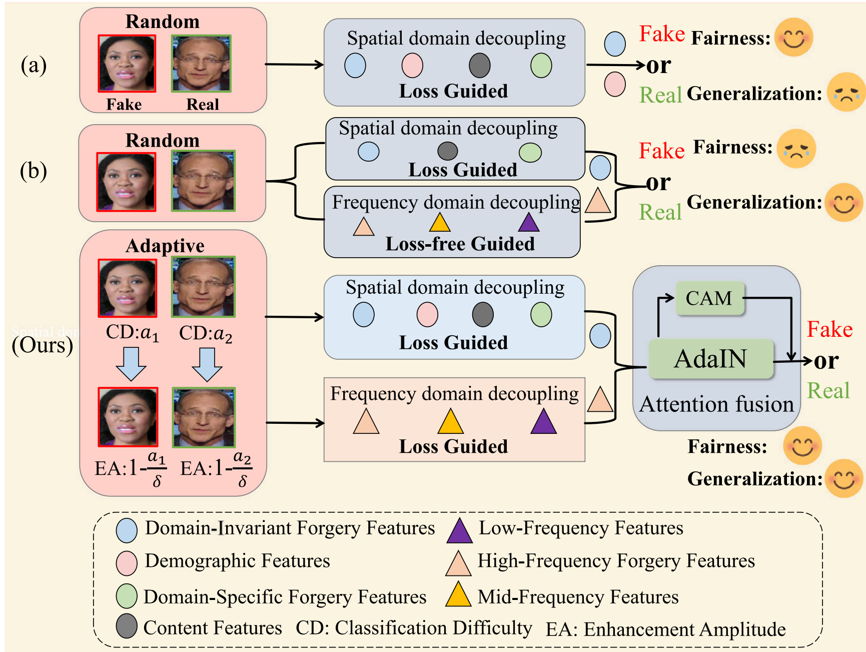

Fig. 1. (a) Exisiting methods based on spatial-domain decoupling. The methods have enhanced fairness, but its generalization is lacking. (b) Exisiting methods based on frequency-domain decoupling. The methods have enhanced generalization, but its fairness is lacking. (Ours) The proposed fair generalization learning framework, achieving improvements in both fairness and generalization compared with previous methods on cross datasets.

suffer from insurmountable limitations due to inherent flaws in their feature learning and domain decoupling strategies. Methods focusing on fairness [11] primarily rely on spatial-domain decoupling, which mitigates bias by separating demographic features from forgery-related representations. However, these methods overemphasize the modeling of demographic information during training, leading to the loss of fine-grained forgery cues and severe generalization degradation—especially when confronted with unseen forgery algorithms. This limitation stems from a critical oversight: spatial-domain features inherently entangle domain-invariant forgery cues, demographic attributes, and content features, and the naive decoupling of demographic features inevitably impairs the integrity of forgery-related representations, as illustrated in Fig. 1(a). On the other hand, methods focusing on generalization [12] adopt frequency-domain decoupling, fusing high-frequency forgery artifacts (robust cues for cross-method detection) with spatial-domain features to enhance generalization performance. Nevertheless, these methods completely fail to address fairness concerns: spatial-domain features still contain uneliminated demographic attributes, and frequency-domain features lack dedicated supervision to prevent bias from entangling with forgery cues, as shown in Fig. 1(b). Furthermore, both categories of methods employ random data augmentation strategies, which introduce noisy or out-of-distribution samples. This leads to overfitting, unstable training, and a further exacerbation of the fairness-generalization trade-off. Specifically, random augmentation fails to prioritize the learning of hard-to-classify samples (e.g., deepfakes of dark-skinned females), thereby widening the performance gap across different demographic groups.

To address the inability to simultaneously achieve fairness and generalization in face forgery detection, we propose a fair generalization learning framework based on cross-domain decoupling and entropy-adaptive enhancement, as shown in Fig. 1(**Ours**). Unlike existing methods that only perform single-domain (spatial or frequency) decoupling or simple feature fusion, our approach employs dedicated fairness and generalization supervision to simultaneously decouple and purify forgery features in both the spatial and frequency domains. It also dynamically adjusts the training process to concurrently mitigate demographic bias and address cross-domain generalization challenges. Specifically, we first design a High-Frequency Feature Decoupling Module (HFFDM) to extract high-frequency forgery features that are less influenced by demographic attributes. To supervise the learning of these features, we design two auxiliary tasks for forgery detection: a frequency-domain feature perception task and a frequency-domain feature contrastive learning task, which provide explicit supervision for learning discriminative and fair frequency-domain features. Second, a spatial-domain decoupling network is utilized to disentangle domain-invariant forgery features, thereby eliminating interference from demographic attributes. We then design a Cross-Domain Attention Fusion Module (CAFM) to fuse the decoupled domain-invariant spatial features with the purified high-frequency features, enabling fair generalization learning under the supervision of a fusion loss function. To further enhance the model's fairness and generalization while addressing the limitations of random augmentation strategies, we introduce an Fairness Enhancement Strategy Based on Entropy-weighted Adaptive (EAFES). To sum up, the main contributions of this paper are summarized as follows:

- We propose a fair generalization framework for face forgery detection by performing cross-domain decoupling of spatial and frequency features. And two auxiliary tasks are introduced to guide the learning of frequency-domain features: frequency-domain perception and contrastive learning.
- We introduce an entropy-adaptive enhancement strategy that adaptively applies data augmentation based on classification difficulty, and incorporate a cross-domain attention fusion module for effective feature fusion.
- Extensive experiments on benchmark deepfake datasets demonstrate that our method consistently surpasses state-of-the-art approaches in achieving fairness-aware and generalizable forgery detection.

2. Related work

2.1. Generalization in face forgery detection

The majority of existing face forgery detection methods fell into the *data-driven* category, including [13]. [14] outline that the unified face attack detection would be a promising research area in the face forensics community. [15] proposed a consistent representation learning framework, aiming to improve the performance of face forgery detection by explicitly constraining the consistency between different representations. [16] proposed an end-to-end reconstruction-classification learning method, which mastered the compact representation of real human faces through reconstruction learning and used the reconstruction differences for forgery recognition. [17] used intra-frame inconsistency to distinguish between real faces and forged faces. Numerous face forgery detection methods [18] were developed, employing various Deep Neural Network (DNN) architectures to enhance the model's ability to detect unseen forgery methods and improve its generalization. To address the generalization issue, disentanglement learning [19] was widely used in forgery detection to extract relevant features while eliminating irrelevant ones. [20] extended this framework by exclusively utilizing common forgery features, which were distinct from specific forgery features. [21] presented feature-level enhancement methods both within and across domains to expand the feature space of the training data and improve generalization. [22] propose a novel Critical Forgery Mining (CFM) framework, which can be flexibly assembled with various backbones to boost their generalization and robustness performance. MoE-FFD [23] innovatively integrate LoRA and Adapter modules with the ViT backbone for face forgery detection, this design enables the model to simultaneously mine global and local forgery clues in a parameterefficient manner. [24] propose a conceptually simple but effective method to efficiently detect forged faces in an image while simultaneously locating the manipulated regions. [25] design a forgery-style mixture module to enhance the diversity of source domains. The model's generalizability is effectively improved by randomly mixing forgery styles. ProDet [26] rethinked the role of blendfake in detecting deepfakes and formulate the process from "real to blendfake to deepfake" to be a progressive transition, and proposed an Oriented Progressive Regularizor(OPR) to establish the constraints that compel the distribution of anchors to be discretely arranged. Furthermore, we introduce feature bridging to facilitate the smooth transition between adjacent anchors. [12] employed a two-stage training process, where high-frequency forgery features decoupled from the frequency-domain were learned alongside spatial-domain features. While this approach enhanced the model's generalization ability, it still exhibited fairness issues due to the entanglement of demographic features in the spatial-domain. Meanwhile, since the data augmentation he adopted was random, this led to a decline in the fairness performance of the model.

2.2. Fairness in face forgery detection

Recent studies mentioned fairness issues in face forgery detection [27]. [9] identified biases in both deepfake datasets and detection models, revealing significant error rate differences across subgroups. [28] assessed the fairness of the MesoInception-4 face forgery detection model on FF++ and found it to be unfair to both genders. [29] conducted a comprehensive analysis of bias in face forgery detection and enriched datasets with diverse annotations to support future research. Additionally, [10] highlighted substantial bias in datasets and detection models, introducing a gender-balanced dataset to mitigate gender-based performance bias. However, this approach yielded only modest improvements and required extensive data annotation. [27] analyzed the problem of racial bias in face forgery detection from the aspects of model, dataset, and evaluation, and contributed a dedicated dataset called the Fair Forgery Detection (FairFD) dataset. [30] proposed a fairness loss to supervise learning across different demographic groups. While it made progress in intra-domain scenarios, its performance remained poor in cross-domain testing. [11] proposed a disentanglement learning framework. Although it disentangled demographic features in the spatial-domain and improved the model's fairness, it suffered from insufficient generalization. DAID [31], for the first time, uncover and formally define a causal relationship between fairness and generalization. Building on the back-door adjustment, DAID show that controlling for confounders (data distribution and model capacity) enables improved generalization via fairness interventions. Previous methods [12] that utilize the combination of frequency domain and spatial domain have enhanced the generalizability of models, while they have all failed to address the issue of fairness—which is the focus of this study. Meanwhile, previous algorithms [11] that account for fairness lack sufficient generalizability due to failing to take into account forgery cues in the frequency domain—and this is also a key focus of our work. Therefore, the key focus of this work is enhancing the fairness and generalization of the model.

3. Proposed method

3.1. Overall framework

We propose a fair generalization learning framework based on cross-domain decoupling and entropy-adaptive enhancement. Xception [32] is adopted as the backbone network for feature extraction. Specifically, the spatial domain decoupling network separates these features into domain-invariant forgery features, domain-specific forgery features, demographic features, and content features, and only inputs the domain-invariant forgery features to the subsequent fusion module. Meanwhile, the high-frequency feature decoupling module (HFFDM) processes the input image through an adaptive high-pass filter (AHPF), extracts high-frequency forgery features, and purifies them through two frequency-domain auxiliary tasks to ensure the discriminative power of the features and the invariance of demographic attributes. To integrate high-frequency forgery features and spatial domain-invariant features, we propose a cross-domain attention fusion module. This module enables effective information fusion while preserving original features. It unifies the two types of features into a robust, fair, and generalized representation for final face forgery classification. The fused features are subsequently passed through the backbone network and used for fairness-aware classification. Additionally, we introduce an fairness enhancement strategy to further improve the model's fairness and generalization capabilities. The overall architecture of the proposed framework is illustrated in Fig. 2. The structure of our decoupled network is consistent with this Fairness [11]. The architecture details of the decoupled network in our proposed method are presented in Fig. 3.

An image pair, comprising one fake and one real image, serves as the input, which is subsequently processed by an encoder built upon the Xception [32] backbone. The encoder decouples domain-specific forgery features f_i^s, domain-invariant forgery features f_i^c, demographic features d_i, and content features c_i. Among them, the domain-invariant forgery features f_i^c serve as the input of our framework.

3.2. High-frequency feature decoupling module

Previous face forgery detection methods based on the frequency domain have improved generalization performance by leveraging high-frequency features, but they still suffer from two key unresolved flaws: (1) They adopt fixed frequency-domain filters, which fail to adaptively adjust to diverse forged artifacts, leading to inadequate extraction of high-frequency features; (2) They lack specific supervision mechanisms for frequency-domain features, resulting in the entanglement of demographic attributes and noise features with high-frequency forgery cues—this ultimately impairs the fairness of detection results. To address the aforementioned issues, we design a High-Frequency Feature Decoupling Module (HFFDM) based on the baseline spatial-domain network. Specifically, we innovatively develop a frequency-domain module that integrates adaptive high-frequency filtering with explicit supervision for fairness and generalization. Built upon the spatial baseline network, HFFDM is designed to extract pure high-frequency forgery features that are invariant to demographic attributes.

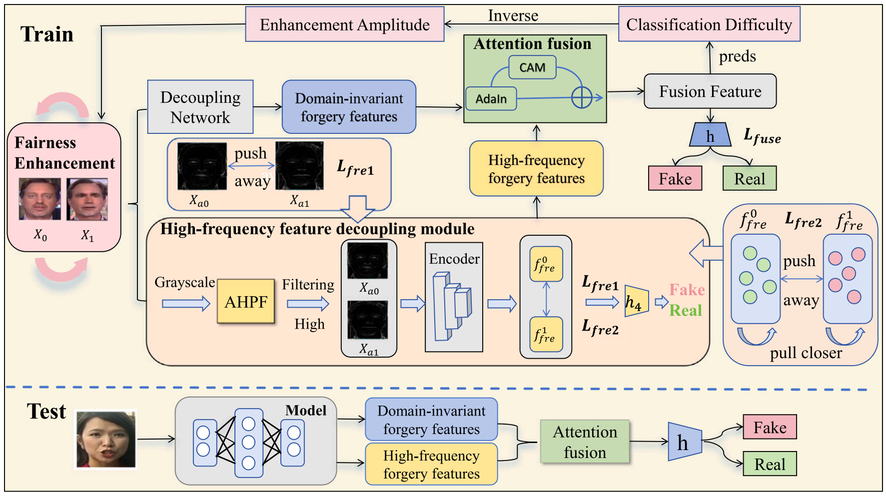

Fig. 2. An overview of our proposed method. (1) For the high-frequency feature decoupling module, we utilize it to expose high-frequency forgery features. Meanwhile, it is trained under the guidance of frequency-domain feature perception loss and frequency-domain feature contrast loss. (2) For the cross-domain attention fusion module, we first fuse the high-frequency forgery features and the domain-invariant forgery features extracted through the spatial-domain decoupling network through an adain module, and then obtain the final fused features by weighting through the channel attention mechanism. (3) For the fairness enhancement strategy based on entropy-weighted adaptive, during training, dynamic enhancement is carried out based on the classification difficulty of the samples, further improving the fairness and generalization of the model.

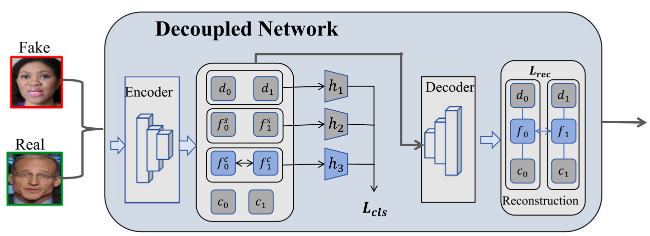

Fig. 3. The architecture details of decoupled network.

Without introducing demographic biases or noise cues, it cooperates with the domain-invariant forgery features in the spatial domain to facilitate fair generalization learning for face forgery detection. In this subsection, we design a HFFDM based on Gaussian filter. To decouple high-frequency forgery features from the frequency-domain, we design an adaptive high-pass filter (AHPF), inspired by the Gaussian filter. The high-pass filter is obtained by subtracting the low-pass Gaussian kernel from the unit impulse kernel.

Finally, the generated high-pass filter kernel is normalized to ensure its central value is -1, and the single-channel high-pass filter kernel is extended to multiple channels to accommodate multi-channel images. This way, each channel shares the same high-pass filter kernel.

Since existing detection methods utilizing frequency cues lack dedicated tasks to supervise decoupled high-frequency forgery features, the model exhibits insufficient generalization. Thus, we design a frequency-domain feature perception task and a frequency-domain feature contrastive learning task for the forgery detection task to supervise the learning of high-frequency forgery features. For these two tasks, we introduce two corresponding losses: frequency-domain feature perception loss and frequency-domain feature contrastive loss.

The frequency domain feature perception task aims to directly classify the extracted high-frequency features as true or false, thereby forcing the high-frequency forged features to have discriminative power, in order to support the generalization performance. It is designed based on cross-entropy loss, with its specific formulation given as follows:

$$\mathcal{L}_{fre1} = H(h_{fre}(f_i^{fre}), G_i), \tag{1}$$

where $\mathcal{L}_{fre1}$ denotes the cross-entropy loss, $\boldsymbol{h}_{fre}$ is the head for the high-frequency forgery features f_i^{fre}, which is implemented by several MLP layers. $\boldsymbol{G}_i \in \{\text{fake}, \text{real}\}$ is the binary classification label.

The task of frequency-domain feature contrastive learning aims to increase the Euclidean distance between the real and fake high-frequency features, while reducing the distance within the same category of features. This further enhances the discriminative power of the frequency-domain features and improves fairness by leveraging the characteristic that the high-frequency features are independent of demographic attributes. The loss form is as follows:

$$\mathcal{L}_{fre2} = \max\left(\left|\boldsymbol{x}_B - \boldsymbol{x}_{P1}\right|_2 - \left|\boldsymbol{x}_B - \boldsymbol{x}_{N1}\right|_2 + b,\ 0\right), \tag{2}$$

where b serves as a margin hyper-parameter. This method minimizes the Euclidean distance between an anchor image ($\boldsymbol{x}_B$) and its similar counterpart ($\boldsymbol{x}_{P1}$), while simultaneously maximizing the gap between the anchor image and its dissimilar counterpart ($\boldsymbol{x}_{N1}$).

The HFFDM aims to extract forgery-related cue information from the frequency-domain perspective. Notably, high-frequency features are less affected by demographic attributes (e.g., race) and thus more fair. Through the combined effect of the AHPF and two frequency-domain losses, HFFDM can extract pure, discriminative, and population-statistical attribute-independent high-frequency forgery features. This innovation effectively addresses the dual drawbacks of fixed filters and the lack of frequency-domain supervision in previous frequency-domain methods. These features provide effective clues for generalization across different forgery methods, while naturally maintaining fairness and complementing the spatial-domain invariant forgery features, laying the foundation for subsequent cross-domain fusion.

3.3. Cross-domain attention fusion module

Traditional fusion strategies for forgery feature representation — such as naive concatenation or static weighted summation — often suffer from limited generalization across diverse forgery techniques; this is largely due to the fact that they fail to dynamically align the statistical characteristics of spatial and frequency domain features, resulting in poor fusion effects and the loss of discriminative cues. To address this, we propose a Cross-domain Attention Fusion Module (CAFM) to fuse frequency-domain forgery information and spatial-domain information, yielding a more informative feature representation. The

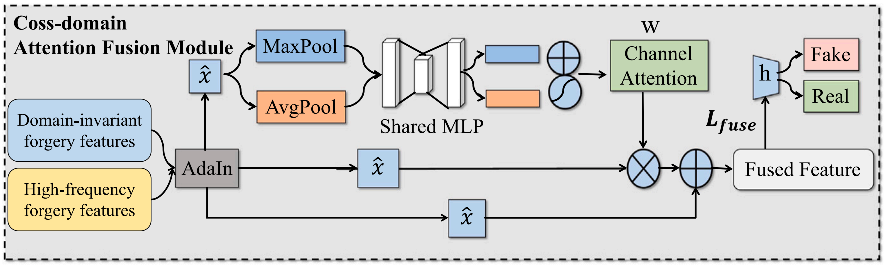

Fig. 4. Cross-domain attention fusion module.

structure diagram is shown in Fig. 4. CAFM takes two input features: the domain-invariant forgery features x from the spatial-domain decoupling network and the high-frequency forgery features y from the HFFDM. For these two features, we first use AdaIN technology to merge the two. The principle of fusion is as follows:first, the input features x and y are normalized, including calculating the mean and standard deviation. Using AdaIN, after the input feature x is normalized, the statistical information of y is used to scale and shift:

$$\hat{x} = \frac{x - \mu_x}{\sigma_x} \cdot \sigma_y + \mu_y. \tag{3}$$

AdaIN adaptively transfers frequency-domain statistics to spatial features through instance normalization, enabling the model to emphasize frequency cues associated with detectable artifacts while preserving spatial details in textured regions.

Within the CAFM, we incorporate a channel attention mechanism to further adjust the channel weights of fused features. This mechanism automatically learns the importance of each channel and assigns a corresponding weight, which reflects the channel's relevance to the final task. Channels with higher weights indicate they contain more critical information — specifically, information associated with forgery — whereas those with lower weights are likely to contain noise or redundant information.

This step completes the initial fusion of feature x and y. The fused feature $\hat{x}$ is then fed into the channel attention module (CAM) to compute the channel attention weights.

$$w = ca(\hat{x}). \tag{4}$$

The fused features are weighted by channel attention weights to generate the final output features.

$$f_i^{fuse} = \hat{x} + \hat{x} \cdot w. \tag{5}$$

Drawing inspiration from prior work [30], the final cross-domain feature fusion loss incorporates a fairness optimization mechanism:

$$\mathcal{L}_{fuse} = \min_{\eta \in \mathbb{R}} \eta + \frac{1}{\alpha|\mathcal{J}|} \sum_{j=1}^{|\mathcal{J}|} [L_j - \eta]_+, \tag{6a}$$

$$\text{s.t. } L_j = \min_{\eta_j \in \mathbb{R}} \eta_j + \frac{1}{\alpha'|\mathcal{J}_j|} \sum_{i:D_i=\mathcal{J}_j} [C(h(f_i^{fuse}), Y_i) - \eta_j]_+. \tag{6b}$$

Here, $|\mathcal{J}|$ represents the size of set $\mathcal{J}$, with each subgroup $\mathcal{J}_j \in \mathcal{J}$, and $|\mathcal{J}_j|$ represents the number of training examples in $\mathcal{J}_j$. $C(\cdot,\cdot)$ is the CE loss; h is the classification head for f_i^{fuse}, sharing the same MLP architecture as other heads, and $\alpha, \alpha' \in (0, 1)$ are two hyperparameters. By constraining the fused features via the fusion loss, we aim to reduce the discrepancies between different racial groups and enhance the fairness of the model.

The design goal of CAFM is to retain the complementary advantages of the two types of features while avoiding the recurrence of population statistical bias. The spatial-domain invariant forged features provide fair underlying spatial cues, and the frequency-domain high-frequency forged features provide high-level pseudo-echo cues with generalization capabilities. Ultimately, a unified feature representation that is both fair and generalizable is generated.

3.4. Fairness enhancement strategy based on entropy-weighted adaptive

Traditional data augmentation usually adopts random augmentation strategies, which have two key drawbacks. First, for hard-to-classify samples, such strategies may introduce noise or out-of-distribution samples, leading to poor model performance on these samples and widening the performance gap across different demographic groups. Second, for easily classified samples, they provide insufficient diversity, which tends to cause model overfitting and limit the model's generalization ability to unknown forgery methods.

Therefore, inspired by [33], we design a fairness enhancement strategy based on entropy-weighted adaptive in the model. Unlike the original entropy-driven enhancement method that only optimizes the classification accuracy, the EAFES proposed in this paper uses entropy values to measure the classification difficulty of each sample and the corresponding demographic group, and adaptively adjusts the enhancement intensity. The structure diagram is shown in Fig. 5. This constitutes a dynamic augmentation strategy. Specifically, within our fairness-oriented augmentation strategy, weaker augmentations are applied to samples with higher classification difficulty—i.e., individuals with darker skin tones—to enhance the model's learning capacity for these cases. Conversely, stronger augmentations are assigned to samples with lower classification difficulty (i.e., individuals with lighter skin tones) to increase sample diversity and thereby improve the model's generalization ability. For our forgery detection model f_θ parameterized by θ and an input sample $\boldsymbol{x} \in \mathbb{R}^n$, $f_\theta(\boldsymbol{x})$ is the network output. For simplicity, we use $g(\boldsymbol{x}) = \mathrm{softmax}(f_\theta(\boldsymbol{x})) \in \mathbb{R}^k$ to denote the output of the softmax function, where k is 2. Therefore, $g(\boldsymbol{x})$ is indeed a probability distribution, i.e., $\sum_{i=1}^k g(\boldsymbol{x})_i = 1$. The classification difficulty of samples is defined as:

$$d(\boldsymbol{x}) = -\sum_{i=1}^{k} g(\boldsymbol{x})_i \log g(\boldsymbol{x})_i, \tag{7}$$

The augmentation magnitude of a sample is inversely proportional to its classification difficulty: the higher the classification difficulty of a sample, the lower its corresponding augmentation magnitude. The enhancement amplitude is defined as follows:

$$mag(\boldsymbol{x}) = 1 + \frac{1}{\log k} \sum_{i=1}^{k} g(\boldsymbol{x})_i \log g(\boldsymbol{x})_i. \tag{8}$$

Unlike traditional augmentation methods, the proposed strategy not only improves the model's generalization but also enhances its fairness. Specifically, we use the entropy value calculated by the model to measure the classification difficulty of samples and adaptively adjust the augmentation magnitude based on this difficulty to enhance the model's fairness. For samples with low classification difficulty, a larger augmentation magnitude is applied to increase diversity and enhance the model's generalization. For samples with high classification difficulty, a smaller augmentation magnitude is applied, enabling the model to focus more on learning these samples.

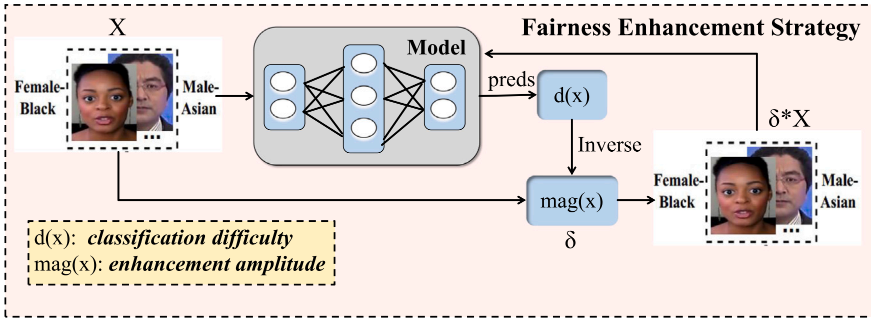

Fig. 5. Fairness enhancement strategy based on entropy-weighted adaptive.

3.5. Overall objective

The overall objective of our framework comprises the spatial-domain entanglement loss, frequency-domain feature perception loss, frequency-domain feature contrastive loss, and cross-domain feature fusion loss.

The spatial-domain entanglement loss is composed of three classification losses for demographic features, domain-specific and domain-invariant forgery features, a contrast losses are used to distinguish between real and fake features in the spatial-domain and a reconstruction loss to ensure consistency between original and reconstructed images at the pixel level. These losses are combined in a weighted sum to create the spatial-domain entanglement loss function for training the framework. The total classification loss is:

$$\mathcal{L}_{cls} = C(h_1(f_i^c), C_i) + \rho_1\, C(h_2(f_i^s), S_i) + \rho_2\, M(h_3(d_i), D_i), \tag{9}$$

where $\mathcal{L}_{cls}$ denotes the cross-entropy loss, $\boldsymbol{h}_1$ is the head for the domain-invariant forgery features, which is implemented by several MLP layers, $\boldsymbol{C}_i \in \{\text{fake}, \text{real}\}$ is the binary classification label. In addition, h_2 is the head for the domain-specific forgery features, $\boldsymbol{S}_i \in \{\text{fake1}, \text{fake2}, \dots, \text{real}\}$ is the multi-classification label, h_3 is the head for the demographic features, $\boldsymbol{D}_i \in \{\text{male-asian}, \text{male-white}, \dots, \text{nonmale-others}\}$ is the multi-classification label. For the classification loss L_{cls}, we configure $\rho_1 = 0.1$ and $\rho_2 = 0.1$.

In the spatial branch, the contrastive regulation loss is formulated mathematically as:

$$\mathcal{L}_{con} = \max\left(\left|\boldsymbol{x}_A - \boldsymbol{x}_P\right|_2 - \left|\boldsymbol{x}_A - \boldsymbol{x}_N\right|_2 + \alpha,\ 0\right), \tag{10}$$

where α is set to 3. This method minimizes the Euclidean distance between an anchor image ($\boldsymbol{x}_A$) and its similar counterpart ($\boldsymbol{x}_P$), while simultaneously maximizing the gap between the anchor image and its dissimilar counterpart ($\boldsymbol{x}_N$).

To preserve the completeness of the extracted features and maintain consistency between the original and reconstructed images at the pixel level, we employ a reconstruction loss. It is formulated as:

$$\mathcal{L}_{rec} = \|X_i - \mathbf{D}(c_i, f_i, d_i)\|_1 + \|X_i - \mathbf{D}(c_i, f_{i'}, d_i)\|_1, \tag{11}$$

In L_{rec} loss, the first term is the self-reconstruction loss, which minimizes reconstruction errors using the latent features of the input image. The second term is the cross-reconstruction loss, which penalizes reconstruction errors by incorporating the partner's forgery feature. These two terms work together to improve feature disentanglement.

The spatial-domain disentanglement loss is as follows:

$$\mathcal{L}_{dis} = \mathcal{L}_{cls} + a_1 \mathcal{L}_{con} + a_2 \mathcal{L}_{rec}, \tag{12}$$

where a_1, a_2 are hyper-parameters for the spatial domain entanglement loss. We set $a_1 = 0.05$ and $a_2 = 0.3$ in Eq. (12).

The final loss function for the training process is formulated as the weighted sum of the aforementioned loss functions.

$$\mathcal{L} = \mathcal{L}_{dis} + \lambda_1 \mathcal{L}_{fre1} + \lambda_2 \mathcal{L}_{fre2} + \lambda_3 \mathcal{L}_{fuse}, \tag{13}$$

where $\lambda_1, \lambda_2, \lambda_3$ are hyper-parameters for balancing the overall loss.

Model Testing. For our method, the inference process of the trained model is illustrated in Fig. 2. In the testing phase, test images are first fed into the trained model. The spatial decoupling network and HFFDM are then adopted to extract domain-invariant forgery features and high-frequency forgery features, respectively. Subsequently, the CAFM fuses these features. Finally, the trained classifier performs forgery detection based on the fused features.

4. Experiment

4.1. Experimental setup

Datasets: To validate the fairness generalization ability of our proposed method, we train our model on the most widely used benchmark FaceForensics++(FF++) [3] and test it on FF++, DeepfakeDetection (DFD) [34], Deepfake Detection Challenge (DFDC) [35], and Celeb-DF [36]. The forged images we use in FF++ are generated by five face manipulation algorithms, including Face2Face (F2F) [37], FaceSwap (FS) [38], DeepFakes(DF) [38], NerualTexture (NT) [39], and FaceShifter(FST) [40]. Since the original datasets do not have the demographic information of each video or image, we follow Ju et al. [30] for data processing, data annotation, and sensitive attributes combination (Intersection). Therefore, the Intersection group contains Male-Asian (M-A), Male-White (M-W), Male-Black (M-B), Male-Others (M-O), Female-Asian (F-A), Female-White (F-W), Female-Black (F-B), and Female-Others (F-O).

Fairness Evaluation Metrics: For detection comparison, the Area Under Curve (AUC) is used to benchmark our approach against previous works, which aligns with the detection evaluation approach adopted in previous works [20]. Regarding fairness, we use four distinct fairness metrics to evaluate the effectiveness of our proposed method. Specifically, we report the Equal False Positive Rate (F_{FPR}) [30], Max Equalized Odds (F_{MEO}) [41], Demographic Parity (F_{DP}) [42] and Overall Accuracy Equality (F_{OAE}) [41], this is consistent with the fairness metrics in this work [11].

Implementation Details: All experiments are implemented using PyTorch and conducted on an NVIDIA RTX 4090 Ti GPU. The FF++ dataset uses the c23 version. During training, we set the batch size to 8, train for 100 epochs, and use the SAM optimizer with a learning rate of $\beta = 3 \times 10^{-4}$. The margin b in Eq. (2) is set to 3. The fairness-related parameters α and α' in Eq. (6) are tuned over the grid {0, 0.1, 0.3, 0.5, 0.7, 0.9}. For the overall loss, we assign the weights $\lambda_1 = 0.05$, $\lambda_2 = 0.1$, and $\lambda_3 = 1.0$ in Eq. (13).

4.2. Results

Performance on Intra-domain sub-datasets: Intra-domain evaluation, conducted on individual forgery sub-datasets, assesses the model's ability to fit each specific domain. As shown in Table 1, our method consistently improves fairness and achieves higher AUC scores on

Table 1
Intra-domain evaluation on FF++. All methods are trained on FF++ and tested on its sub-datasets generated by five forgery methods (*e.g.*, F2F denotes Face2Face [37]). Lower fairness metric values indicate better fairness, and higher AUC indicates better detection performance.

Testing set	Method	Fairness metrics (%, ↓)				Detection metric (%, ↑)
		F_{FPR}	F_{MEO}	F_{DP}	F_{OAE}	AUC
F2F	DAW-FDD [30]	20.42	12.66	35.46	11.58	97.74
	Fairness [11]	17.42	10.00	**33.20**	9.56	98.65
	Ours	**16.00**	**6.49**	39.77	**5.56**	**99.10**
FS	DAW-FDD [30]	32.96	14.52	21.39	**3.95**	98.62
	Fairness [11]	26.32	9.97	**19.30**	6.70	99.23
	Ours	**18.00**	**6.93**	22.74	5.78	**99.40**
NT	DAW-FDD [30]	23.64	20.83	20.50	17.36	94.99
	Fairness [11]	23.98	16.83	16.03	13.61	**96.35**
	Ours	**16.20**	**15.08**	**7.36**	**9.53**	96.30
DF	DAW-FDD [30]	20.41	12.66	9.99	6.16	98.20
	Fairness [11]	17.42	9.02	**9.43**	5.86	99.05
	Ours	**16.00**	**6.49**	16.22	**3.77**	**99.50**
FST	DAW-FDD [30]	25.36	10.05	10.34	8.79	98.02
	Fairness [11]	15.38	7.79	**6.45**	5.70	98.96
	Ours	**12.00**	**6.49**	9.88	**2.63**	**99.30**

Table 2
Comparison with different methods in terms of improving fairness and detection generalization under both intra-domain (FF++) and cross-domain (DFDC, Celeb-DF, and DFD) scenarios. ↑ means higher is better and ↓ means lower is better.

Dataset	Method	Xception [32]					ResNet-50 [43]					EfficientNet-B3 [44]				
		Fairness metrics (%)↓				Detection metric (%)↑	Fairness metrics (%)↓				Detection metric (%)↑	Fairness metrics (%)↓				Detection metric (%)↑
		F_{FPR}	F_{MEO}	F_{DP}	F_{OAE}	AUC	F_{FPR}	F_{MEO}	F_{DP}	F_{OAE}	AUC	F_{FPR}	F_{MEO}	F_{DP}	F_{OAE}	AUC
FF++	DAW-FDD [30]	14.06	10.55	10.97	8.72	97.46	30.36	9.74	8.89	7.42	93.23	23.33	26.15	24.74	21.23	94.92
	UCF [20]	21.52	13.06	15.06	10.58	97.10	35.13	10.87	10.81	8.05	95.92	20.92	33.08	30.01	24.56	94.21
	DFS-GDD [12]	15.72	12.35	13.60	9.45	97.85	32.15	10.70	10.55	7.84	97.75	22.40	28.32	22.50	22.60	96.40
	Fairness [11]	10.63	8.15	10.41	7.60	98.28	22.70	**9.28**	8.72	5.74	97.72	11.19	20.61	18.40	16.18	95.39
	Ours	**7.60**	**7.71**	**9.25**	**7.01**	**98.70**	**18.60**	10.43	**5.06**	**5.68**	**97.80**	**9.20**	**11.68**	**11.57**	**9.35**	**98.30**
DFDC	DAW-FDD [30]	45.14	35.77	18.59	14.07	59.96	44.07	34.14	18.72	24.58	60.11	50.73	43.79	18.31	29.57	58.29
	UCF [20]	53.07	44.44	15.70	23.22	60.03	43.39	35.62	15.86	19.15	61.06	42.79	40.54	19.35	21.13	58.85
	DFS-GDD [12]	39.30	**26.07**	16.43	18.18	58.70	40.25	30.46	14.60	25.32	60.80	36.64	39.26	19.70	27.82	60.30
	Fairness [11]	40.73	34.48	9.69	**13.71**	61.47	37.17	27.78	10.94	**18.52**	59.76	22.89	33.78	12.35	**16.73**	60.67
	Ours	**25.8**	43.60	**9.29**	18.66	**63.90**	**23.5**	**24.29**	**8.97**	24.48	**63.20**	**15.80**	**29.55**	**11.52**	26.04	**61.90**
Celeb-DF	DAW-FDD [30]	22.31	20.60	**11.65**	49.71	69.55	26.82	21.93	20.80	47.14	75.70	31.36	21.79	6.91	50.86	70.14
	UCF [20]	27.81	25.96	16.51	48.63	71.73	32.17	28.28	19.38	45.15	76.44	24.95	22.41	15.14	58.48	72.65
	DFS-GDD [12]	38.50	47.83	35.03	21.20	67.50	30.47	25.50	19.28	40.36	78.61	23.80	27.15	23.75	56.80	76.43
	Fairness [11]	10.62	**12.77**	15.04	36.01	74.42	**11.55**	17.01	17.21	29.58	78.55	13.00	**9.73**	**5.21**	55.74	75.32
	Ours	**4.00**	27.95	31.43	**20.57**	**80.70**	18.6	**16.14**	**17.18**	**14.54**	**79.60**	**8.50**	26.30	22.97	**15.16**	**78.30**
DFD	DAW-FDD [30]	34.02	29.37	15.75	11.31	71.42	33.05	24.24	7.12	27.08	77.05	32.72	28.74	17.12	24.70	74.76
	UCF [20]	42.66	33.41	20.24	19.84	81.88	42.54	33.17	5.24	30.98	78.97	36.59	27.32	25.83	9.36	76.76
	DFS-GDD [12]	40.10	29.50	11.73	7.96	75.20	35.33	31.62	15.50	28.76	79.92	34.80	27.25	23.10	9.45	77.56
	Fairness [11]	26.08	21.37	**11.65**	8.37	84.82	25.71	20.02	**2.34**	25.60	79.67	29.34	24.52	**11.46**	5.11	77.28
	Ours	**7.90**	**6.34**	20.94	**3.65**	**85.90**	**16.90**	**15.10**	19.65	**10.06**	**85.70**	**23.70**	**19.52**	21.25	**1.06**	**86.80**

nearly all sub-datasets compared to both DAW-FDD [30] and the Fairness [11] model. These results demonstrate the effectiveness of our approach in mitigating domain-specific biases.

Performance of Fairness Generalization: We compared with the fairness-based method Fairness, DAW-FDD, and the frequency-domain method DFS-GDD [12]. The results are shown in Table 2. Taking Xception backbone as an example, our method has superior fairness generalization ability compared to other methods, while simultaneously achieving the best detection results. Specifically, compared to this Fairness, our method has a 14.93% reduction in F_{FPR} on DFDC and drops F_{FPR} by 6.62% on Celeb-DF, 18.18% on DFD compared with Fairness. In addition, compared to this Fairness and DAW-FDD, our method has achieved improvements in AUC metrics in all four data sets, demonstrating the generalization of our method. Specifically, compared to the Fairness, our method achieves a 2.43% improvement in AUC on DFDC, a 6.28% improvement on Celeb-DF, and a 1.08% improvement on DFD. Given the potential conflicts among fairness metrics, certain results may be suboptimal. Consequently, devising reasonable fairness evaluation metrics remains an important consideration for future work. Overall, our method yields improvements in both fairness and generalization.

4.3. Ablation study

Effectiveness of Fairness Enhancement Strategy: The results in the first two rows of Table 3 validate the efficacy of our fairness enhancement strategy. Compared with the baseline spatial-domain decoupling network, the fairness enhancement strategy has a 11.53% reduction in F_{FPR} on DFDC and has a 3.98% improvement in AUC on Celeb-DF. The results indicate that the enhancement strategy further strengthens the model's generalization and fairness.

Effectiveness of High-Frequency Feature Decoupling Module: As observed from the results in the second and third rows of Table 3, the model's generalization is further improved overall from the perspective of the AUC metric. Meanwhile, this method has a 11.9% reduction in F_{FPR} on DFD. This module assists spatially disentangled domain-invariant forgery features by extracting high-frequency forgery features: the latter can supplement forgery cues in the spatial domain, thereby enhancing generalizability; The former improves fairness by disentangling demographic features. The combination of the two achieves the effect of balancing both fairness and generalizability. As observed from the results in the last three rows of Table 3, our frequency-domain feature perception loss (*e.g.*, F_{FPR} drops 3.8% and the AUC increases 0.9% on Celeb-DF) and frequency-domain feature contrastive loss (*e.g.*, F_{FPR} drops 2.8% on DFDC and the AUC increases 0.9% on Celeb-DF). This indicates that these two losses further enhance the model's generalization and fairness.

Effectiveness of Cross-Domain Attention Fusion Module: As observed from the results in the third and fourth rows of Table 3, the proposed method exhibits a reduction across all four fairness metrics while achieving an improvement in the AUC metric. This module

Table 3
Ablation study of the proposed framework, evaluating the effectiveness of the fairness enhancement strategy, high-frequency feature decoupling, cross-domain attention fusion module, and the two frequency losses $\mathcal{L}_{fre1}$ and $\mathcal{L}_{fre2}$. All models are trained on FF++ only.

Module configuration					FF++		DFDC		Celeb-DF		DFD	
Enhance	Freq.	Fusion	$\mathcal{L}_{fre1}$	$\mathcal{L}_{fre2}$	$F_{FPR}\downarrow$	AUC↑	$F_{FPR}\downarrow$	AUC↑	$F_{FPR}\downarrow$	AUC↑	$F_{FPR}\downarrow$	AUC↑
✗	✗	✗	✗	✗	10.63	98.28	40.73	61.47	10.62	74.42	26.08	84.82
✓	✗	✗	✗	✗	7.80	98.40	29.20	62.20	11.10	78.40	22.70	82.60
✓	✓	✗	✗	✗	10.40	98.40	36.50	63.90	10.00	78.30	10.80	85.70
✓	✓	✓	✗	✗	7.90	98.60	30.30	63.40	7.90	78.70	7.90	87.00
✓	✓	✓	✓	✗	7.80	98.60	28.60	63.80	4.10	79.60	8.90	**88.00**
✓	✓	✓	✓	✓	**7.60**	**98.70**	**25.80**	**63.90**	**4.00**	**80.70**	**7.90**	85.90

Table 4
Performance comparison of different backbones with and without our method on four deepfake datasets.

Model	FF++			Celeb-DF			DFD			DFDC		
	F_{FPR}	AUC	EER	F_{FPR}	AUC	EER	F_{FPR}	AUC	EER	F_{FPR}	AUC	EER
Xception	31.31	92.77	16.72	27.55	62.66	45.45	35.14	74.34	34.78	52.77	56.72	50.25
Xception+Ours	**7.60**	**98.70**	**8.90**	**4.00**	**80.70**	**27.32**	**7.90**	**85.90**	**22.25**	**25.80**	**63.90**	**44.30**
ResNet-50	34.69	94.83	13.85	24.94	70.64	38.23	31.76	76.02	35.40	45.84	58.08	49.70
ResNet-50+Ours	**18.60**	**97.80**	**10.25**	**18.60**	**79.60**	**28.76**	**16.90**	**85.70**	**22.60**	**23.50**	**63.20**	**45.50**
EN-B3	18.78	93.55	14.50	30.86	62.36	46.21	39.37	75.87	32.78	62.38	57.81	50.10
EN-B3+Ours	**9.20**	**98.30**	**9.25**	**8.50**	**78.30**	**29.80**	**23.70**	**86.80**	**22.40**	**15.80**	**61.90**	**47.65**
ConvNet	16.23	95.68	12.76	22.34	72.23	35.50	28.50	78.60	28.74	40.50	59.55	46.72
ConvNet+Ours	**7.50**	**99.20**	**8.40**	**6.20**	**82.90**	**24.20**	**7.80**	**88.20**	**19.85**	**26.30**	**68.30**	**39.36**

Table 5
Robustness evaluation on FF++ under different perturbation types. Higher values indicate better robustness (%).

Method	Saturation	Contrast	Block	Noise	Blur	Pixel	Avg
DAW-FDD [30]	97.30	97.24	97.40	78.50	88.60	90.80	91.64
Fairness [11]	98.20	98.16	98.22	82.30	90.90	93.80	93.60
Ours	**98.50**	**98.45**	**98.60**	**84.45**	**95.40**	**97.60**	**95.50**

fuses domain-invariant forgery features in the spatial domain and high-frequency forgery features in the frequency domain via AdaIN, and incorporates a spatial attention mechanism to adjust channel weights. Specifically, it assigns higher attention weights to channels relevant to forgery features, thereby enhancing the detection capability of the fused features. This indicates that the fusion module further enhances the model's generalization and fairness.

4.4. Comparison with state-of-the-art methods

Fairness Generalization Performance of Different Backbones: To examine the fairness generalization capability of our proposed method concerning backbone selection, we substitute the Xception backbone with ResNet-50 [43], EfficientNet-B3 [44] and ConvNet [45]. The results in Table 4 indicate that our method based on different backbones shows similar superior results. Such outcomes suggest that our proposed approach is not limited to backbone choice, but is effective and applicable to diverse backbone settings.

Robustness evaluation on FF++: Since deepfake videos are easily altered on various social platforms, the robustness of ours against some common perturbations is investigated. We compare our method with DAW-FDD and Fairness methods in frame-level AUC. Following the same settings of [46], we conducted robustness evaluation on the FF++ dataset under various perturbations, which includes six different types of perturbations: color saturation, color contrast, block wise, gaussian noise, gaussian blur, pixelation. The results in Table 5 indicate that our method is more robust against various perturbations compared with previous methods.

Complexity analysis: We conducted a complexity evaluation of our method against the Fairness method on the Xception backbone network. As shown in Table 6, Our method consistently improves AUC across all four datasets, with only marginal increases in parameters and FLOPs, which remain well within practical limits.

Stability Evaluation: The stability comparison of Fairness with ours over 5 random runs is shown in Table 7. Our method shows superior fairness and detection mean score out of 5 random runs compared to Fairness. This suggests that our approach has a robust and formidable capacity to improve fairness.

Parameter sensitivity analysis: To validate the rationality of empirically tuned hyperparameters and investigate the impact of core hyperparameters on the model's fairness and generalization performance, we conduct a comprehensive parameter sensitivity analysis on the *FF++* benchmark dataset (the primary training dataset). We select three categories of key hyperparameters for evaluation: (1) the kernel size ($kernel\_size$) and Gaussian standard deviation (σ) of the AHPF in the HFFDM, the experimental results are shown in Table 8; (2) the loss weight coefficients λ_1, λ_2, and λ_3 in the overall objective function that balance the frequency-domain feature perception loss ($\mathcal{L}_{fre1}$), frequency-domain feature contrastive loss ($\mathcal{L}_{fre2}$), and cross-domain feature fusion loss ($\mathcal{L}_{fuse}$), respectively. The experimental results are shown in Table 9. (3) the contrastive margins b and α in the frequency-domain feature contrastive loss ($\mathcal{L}_{fre2}$) and contrastive regulation loss ($\mathcal{L}_{con}$), respectively. The experimental results are shown in Table 10; For all sensitivity experiments, we report the F_{FPR} as the core fairness metric and AUC as the detection performance metric to quantify the model's behavior with varying hyperparameters.

Compared with State-of-the-arts: We conducted a comparison with recent state-of-the-art methods, including LSDA [21], ProDet [26] and the fairness detection methods DAID [31], BPFA [27]. All methods are trained on the FF++ c23 set and evaluated across different datasets. Since other methods did not reproduce on the fairness metric, we conducted cross-domain evaluation experiments under the dataset of fairness labels. As can be seen from the results in Table 11, our method is more advantageous in both fairness and generalizability.

Table 6
Complexity analysis in Frame-level AUC.

Method	Params (M)	FLOPs (G)	AUC (FF++)	AUC (DFDC)	AUC (CDF)	AUC (DFD)
Fairness [11]	**24.01**	**18.38**	98.28	61.47	74.42	84.82
Ours	25.88	19.48	**98.70**	**63.90**	**80.70**	**85.90**

Table 7
Detection mean and standard deviation (in parentheses) on cross-domain testing sets across 5 experimental repeats. Each method is trained only on FF++. (mean ± std over 5 runs).

Datasets	Methods	F_{FPR}↓	F_{OAE}↓	AUC↑	EER↓	AP↑
DFD	Fairness	25.15 ± 2.02(+)	8.35 ± 1.25(+)	82.85 ± 2.60(+)	24.65 ± 2.45(+)	87.25 ± 1.05(+)
	Ours	**8.65** ± 1.22	**4.23** ± 1.12	**85.80** ± 2.26	**21.10** ± 2.34	**90.50** ± 0.75
Celeb-DF	Fairness	10.56 ± 3.86(+)	33.23 ± 6.25(+)	75.10 ± 1.85(+)	33.25 ± 3.42(+)	79.10 ± 1.64(+)
	Ours	**6.45** ± 2.65	**22.55** ± 5.70	**79.05** ± 1.75	**28.32** ± 2.78	**84.60** ± 1.56
DFDC	Fairness	38.30 ± 4.53(+)	**18.42** ± 1.72(−)	62.15 ± 0.70(+)	45.75 ± 5.23(+)	67.34 ± 2.21(+)
	Ours	**26.52** ± 3.58	20.55 ± 2.15	**63.68** ± 0.64	**44.50** ± 4.70	**68.50** ± 2.52
+/−		**3/0**	**2/1**	**3/0**	**3/0**	**3/0**

Table 8
Parameter sensitivity analysis of σ and kernel size on different datasets. ↓: lower is better, ↑: higher is better.

Hyper-parameters		FF++			Celeb-DF			DFD		
σ	Kernel size	F_{FPR} ↓	AUC ↑	EER ↓	F_{FPR} ↓	AUC ↑	EER ↓	F_{FPR} ↓	AUC ↑	EER ↓
1.0	1.0	8.80	98.45	9.25	6.27	79.86	28.32	8.58	83.34	25.45
1.0	2.0	8.15	98.58	9.10	5.40	80.35	27.45	8.21	84.36	23.58
1.0	**3.0**	**7.60**	**98.70**	**8.90**	**4.00**	**80.70**	**27.32**	**7.90**	**85.90**	**22.25**
2.0	3.0	7.26	98.56	9.15	5.34	80.28	27.89	8.32	85.23	23.50
3.0	3.0	8.45	98.40	10.20	6.41	79.62	29.40	9.22	84.12	25.55

Table 9
Sensitivity analysis of loss weight coefficients on different datasets. ↓: lower is better, ↑: higher is better.

Hyper-parameters			FF++		Celeb-DF		DFD	
λ_1	λ_2	λ_3	F_{FPR} ↓	AUC ↑	F_{FPR} ↓	AUC ↑	F_{FPR} ↓	AUC ↑
0.07	0.3	1.0	8.76	98.45	6.05	78.35	8.55	82.10
0.06	0.2	1.0	8.25	98.50	5.62	79.60	8.28	84.45
0.05	**0.1**	**1.0**	**7.60**	**98.70**	**4.00**	**80.70**	**7.90**	**85.90**
0.04	0.1	1.0	7.84	98.67	4.87	80.46	8.12	85.37
0.05	0.1	0.8	7.42	98.50	5.24	80.18	8.35	84.90
0.05	0.1	1.2	7.55	98.32	6.46	79.35	9.20	83.24

Table 10
Parameter sensitivity analysis of b and α on different datasets. ↓: lower is better, ↑: higher is better.

Hyper-parameters		FF++			Celeb-DF			DFD		
b	α	F_{FPR} ↓	AUC ↑	EER ↓	F_{FPR} ↓	AUC ↑	EER ↓	F_{FPR} ↓	AUC ↑	EER ↓
2.0	3.0	7.25	98.56	10.25	5.48	80.56	29.50	8.16	84.52	25.40
3.0	**3.0**	**7.60**	**98.70**	**8.90**	**4.00**	**80.70**	**27.32**	**7.90**	**85.90**	**22.25**
3.0	2.0	7.36	98.62	9.80	4.82	80.25	27.76	8.35	85.20	22.87
3.0	4.0	8.22	98.50	10.50	5.60	79.68	28.25	9.25	84.22	23.34

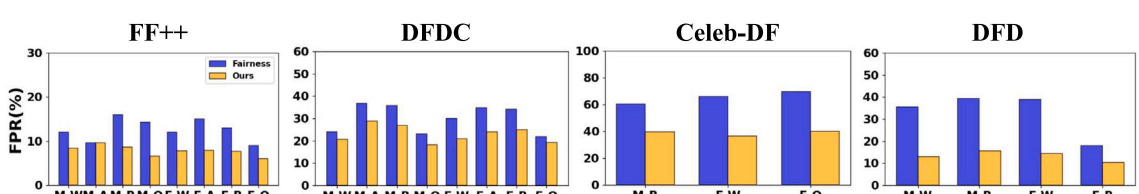

Fig. 6. Comparison of FPR on Intersectional subgroups. Models are trained on FF++ and tested on FF++, DFDC, Celeb-DF, and DFD. The subgroups not represented in Celeb-DF and DFD are inapplicable.

4.5. Visualization

Comparison on Intersectional Subgroups: We present detailed results of the False Positive Rate (FPR) on each subgroup across all datasets, as shown in Fig. 6. The results clearly indicate that our approach significantly narrows the disparity between these subgroups, *e.g.*, the maximum FPR gap of Fairness on Celeb-DF is 9.3%, while

Table 11
Comparison with different methods in terms of improving detection generalization under both intra-domain (FF++) and cross-domain (DFDC, Celeb-DF, and DFD) scenarios. ↑ indicates higher is better, while ↓ indicates lower is better.

Method	Celeb-DF					DFD					DFDC				
	Fairness metrics (%) ↓				Detection metric (%) ↑	Fairness metrics (%) ↓				Detection metric (%) ↑	Fairness metrics (%) ↓				Detection metric (%) ↑
	F_{FPR}	F_{MEO}	F_{DP}	F_{OAE}	AUC	F_{FPR}	F_{MEO}	F_{DP}	F_{OAE}	AUC	F_{FPR}	F_{MEO}	F_{DP}	F_{OAE}	AUC
CORE [15]	32.30	31.58	22.15	43.58	65.10	36.00	23.26	22.51	7.53	81.10	52.20	38.76	22.86	15.17	61.10
RECCE [16]	33.80	33.41	18.65	40.45	70.80	43.00	34.79	31.53	6.45	80.50	50.90	29.25	25.34	15.63	59.40
UIA-ViT [17]	21.60	28.17	20.02	32.39	75.40	19.00	32.25	25.86	24.02	78.80	44.90	62.04	15.83	20.43	66.20
DFS-GDD [12]	38.50	47.83	35.03	21.20	67.50	40.10	29.50	11.73	7.96	75.20	39.30	**26.07**	16.43	18.18	58.70
LSDA [21]	17.40	17.20	26.79	24.77	76.80	26.50	35.10	32.50	26.00	84.10	57.00	77.80	15.88	39.02	60.90
ProDet [26]	15.25	16.54	24.34	22.35	76.54	25.76	33.80	31.65	25.22	83.30	58.43	45.80	15.23	36.05	62.45
BPFA [27]	8.00	**12.60**	16.20	27.60	82.70	12.60	14.30	15.80	**6.80**	85.80	33.50	35.80	9.60	**11.30**	67.50
DAID [31]	8.23	14.44	16.15	25.75	75.80	12.55	15.60	13.70	16.02	82.65	32.45	35.25	10.24	14.02	63.60
Ours (ConvNet)	**6.20**	15.30	**14.30**	**22.80**	**82.90**	**7.80**	**10.80**	**11.80**	7.50	**88.20**	**26.30**	36.70	**9.50**	14.00	**68.30**

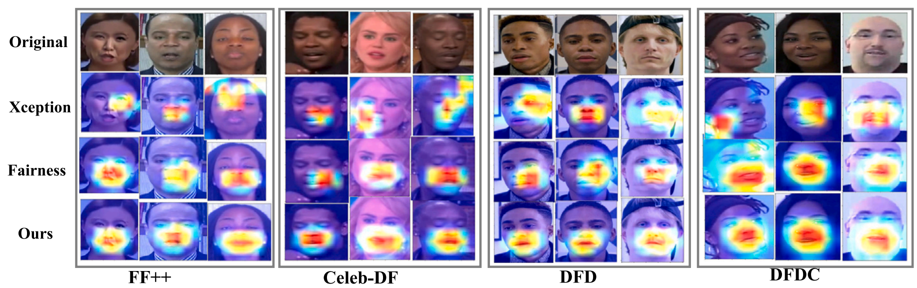

Fig. 7. Grad-CAM visualizations of the Original (first row), Xception (second row), Fairness (third row), and Ours (fourth row) on the intra-domain dataset (FF++) and cross-domain datasets (DFDC, Celeb-DF, and DFD).

our method lowers the gap to 3.73%. Overall, our method achieves a consistent and significant reduction in FPR across all test datasets, which underscores the model's fairness.

Visualization of the Saliency Map: To more intuitively demonstrate the effectiveness of our method, we visualize the Grad-CAM [47] of Xception [32], Fairness [11] and our method, respectively, as shown in Fig. 7. As can be seen from the visualization results, models based on Xception focus more on the edge information of images and thus exhibit poor generalizability. Grad-CAM visualizations reveal that while the Fairness model, which decouples demographic attributes in the spatial domain, demonstrates improved fairness in cross-domain scenarios, it tends to overfit to small local regions or focus on irrelevant noise outside the facial area when confronted with unseen forgery methods—especially for faces with darker skin tones. In contrast, across diverse forgery methods and demographic groups, the activation regions of our proposed method consistently demonstrate that the model focuses on facial salient features, thereby reflecting its fair generalization ability.

UMAP Visualization of High-Frequency Features: Theoretically, the demographic invariance of high-frequency features is grounded in the distinct ways facial attributes and forgery artifacts are encoded in the frequency domain. Demographic attributes, such as race and gender, are predominantly reflected in global facial structures and spatial color distributions (e.g., facial contour, bone structure, and skin pigmentation), which naturally reside in the low-frequency spectrum. Conversely, the high-frequency domain primarily captures fine-grained textures, sharp edges, and, crucially, the microscopic noises and blending artifacts inherently introduced during the deepfake generation process. Therefore, by extracting high-frequency cues, our method intrinsically filters out the low-frequency demographic confounders while preserving the essential patterns required for forgery detection.

To empirically validate this theoretical assumption and examine whether our high-frequency representations inadvertently encode demographic information, we conduct a visualization analysis in the frequency domain. We project the extracted high-frequency forgery features into a 2D space using UMAP [48], as illustrated in Fig. 8. In the projected space, samples from different races and genders are highly intermixed, showing no discernible subgroup clustering or distinct decision boundaries. This observation confirms that the learned high-frequency representations exhibit extremely low demographic sensitivity. Together with our theoretical justification, these empirical

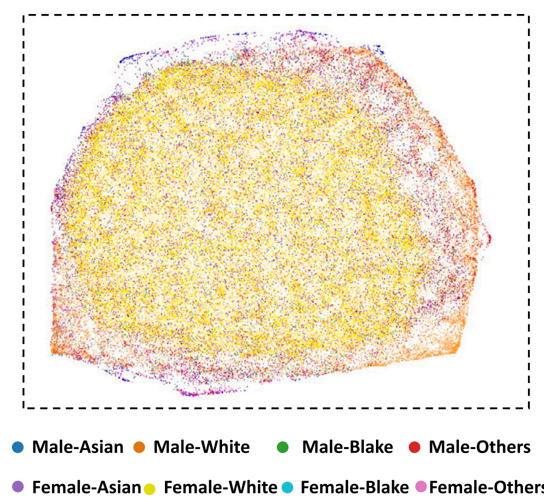

Fig. 8. UMAP visualization of high-frequency features on FF++.

results strongly support the use of high-frequency representations as a principled approach for fairness-aware generalization.

5. Conclusion

In this paper, we propose a fair generalization learning framework to jointly address the challenges of fairness and generalization in face forgery detection. Our approach introduces a cross-domain decoupling mechanism that disentangles domain-invariant features in the spatial-domain and high-frequency forgery features in the frequency-domain. To facilitate effective learning of high-frequency representations, we design two auxiliary tasks: frequency-domain feature perception and frequency-domain contrastive learning. Furthermore, we incorporate a cross-domain attention fusion module to integrate complementary features and perform fairness-aware generalization learning under a unified supervision scheme. To further enhance fairness and generalization during training, we propose an entropy-weighted adaptive enhancement strategy that dynamically adjusts data augmentation based on classification difficulty. Extensive experiments conducted on multiple benchmark face forgery datasets validate the effectiveness of our method, demonstrating consistent improvements over state-of-the-art approaches in terms of both fairness and generalization. The limitation of this work is its reliance on a CNN-based backbone. In future research, we plan to explore this line of inquiry further by incorporating more advanced backbone architectures, such as Vision Transformers (ViT) and CLIP. Furthermore, extending our algorithm to detect other forgery techniques, like those based on diffusion models, or integrating additional facial attributes (e.g., age, expression) presents an interesting direction for future investigation.

CRediT authorship contribution statement

Fan Cheng: Methodology, Conceptualization. **Linkai Tian:** Writing – original draft. **Fanjun Meng:** Funding acquisition, Formal analysis. **Xianliang Wang:** Funding acquisition, Formal analysis. **Mingsha Peng:** Resources, Funding acquisition, Formal analysis. **Liangliang Su:** Visualization, Validation. **Zhize Wu:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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